

Localizing the Lesion

Presenter Cheat Sheet -- 5 Cases

CASE 1: Severe Shoulder Pain Followed by Weakness

Parsonage-Turner Syndrome (Neuralgic Amyotrophy)

35M painter | No trauma | Viral illness 3 weeks ago | Pain preceded weakness by 10 days

DIFFERENTIALS & RULE-OUTS

- C5-C6 Radiculopathy: paraspinals normal + patchy pattern (not single root)
- Upper Trunk Plexopathy: trunk = all muscles in that division; PTS = patchy individual nerves
- Rotator Cuff Tear: no neurogenic EMG findings in a tear
- Mononeuritis Multiplex: vasculitis + systemic sx; PTS = viral prodrome + shoulder predilection

PHYSICAL EXAM -- WHY

- Deltoid 2/5: axillary nerve inflamed
- Infraspinatus 2/5: suprascapular nerve inflamed
- Biceps 3/5: musculocutaneous nerve inflamed (less severe = patchy)
- **Hand 5/5: C8-T1 spared -- PTS hits shoulder girdle, not hand**
- Biceps reflex 1+: C5-C6 efferent limb weakened; Triceps 2+: C7 spared
- Neg Spurling's: no radiculopathy (PTS = plexus/nerve level)

NCS -- WHY

- Median/Ulnar/Radial SNAPs normal: these nerves are NOT affected

LABC 5 uV (ABNORMAL): sensory branch of musculocutaneous. Abnormal = post-ganglionic. If C6 root, LABC would be normal (DRG intact).

- Axillary motor 1.5 mV: axonal loss in axillary nerve
- Musculocutaneous motor 2.8 mV: partial axonal loss
- Median/Ulnar motor normal: C8-T1 territory intact

EMG -- WHY

- Deltoid 3+ fibs: axillary nerve -- orphaned fibers fibrillate
- Infraspinatus 3+ fibs: suprascapular nerve
- Serratus Anterior 2+ fibs: long thoracic nerve (most commonly hit in PTS)
- Biceps 2+ fibs: musculocutaneous

Pronator Teres NORMAL: C6-C7 median muscle SPARED = patchy (not root pattern)

C5-C6 Paraspinals NORMAL: post-ganglionic. Root lesion would show paraspinal fibs.

30% bilateral. 70-80% recover by 2-3 years. AIN variant exists.

CASE 2: Hand Weakness After a Fall

Klumpke's Palsy (Lower Trunk Plexopathy)

24M bike messenger | Grabbed tree branch = upward traction | 4 weeks out, progressive clawing

DIFFERENTIALS & RULE-OUTS

- C8-T1 Radiculopathy: MAC + Ulnar SNAPs abnormal (post-ganglionic). In radic, SNAPs = normal.
- Combined Ulnar + Median Neuropathy: doesn't explain MAC territory loss or EIP (radial) weakness
- TOS: chronic/positional, not acute traction
- T1 Root Avulsion: NO Horner's = not T1 root

PHYSICAL EXAM -- WHY

Thenar AND hypthenar atrophy: crosses median/ulnar boundary = proximal to where they split

- Claw hand: lumbricals/interossei (ulnar) weak = MCP hyperextension + IP flexion
- APB/FDI/ADM 2/5: all C8-T1 from different nerves
- Triceps/Deltoid 5/5: C5-C7 upper trunk spared

NO Horner's: sympathetic chain at T1 root intact = post-ganglionic trunk lesion

NCS -- WHY

- Median sensory (index) normal: C6-C7 lateral cord territory
- Ulnar sensory 5 uV: C8-T1 lower trunk damaged, Wallerian degeneration

MAC 4 uV (ABNORMAL): THE MONEY TEST. Abnormal = post-ganglionic. In C8 radic, MAC = normal (DRG intact).

- Both Median motor (APB) 3.5 + Ulnar motor (ADM) 2.5 LOW: C8-T1 fibers through lower trunk

EMG -- WHY

- APB + FDI + ADM all 2+ fibs: C8-T1 across median + ulnar nerves
- EIP 1+ fibs: radial C7-C8 -- C8 fibers pass through lower trunk
- Biceps/Deltoid normal: upper trunk spared

C8-T1 Paraspinals NORMAL: post-ganglionic (trunk, not root)

CASE 3: Flail Arm After Motorcycle Accident

Traumatic C5-T1 Root Avulsions

62M construction worker | MVA + clavicle fracture | 6 months out | Complete arm paralysis

DIFFERENTIALS & RULE-OUTS

- Cord injury: would be bilateral or have long tract signs (legs)
- MMN: distal-predominant + purely motor; this has sensory loss
- Diabetic polyneuropathy: symmetric stocking-glove; this is unilateral

PHYSICAL EXAM -- WHY

- 0-1/5 throughout: ALL motor axons C5-T1 destroyed at root level

Horner's (ptosis, miosis, anhidrosis): T1 sympathetic fibers avulsed BEFORE leaving spinal canal

- Winged scapula: long thoracic (C5-C7) roots all avulsed
- All reflexes absent: both afferent + efferent limbs disrupted

NCS -- THE PATHOGNOMONIC PATTERN

ALL SNAPs PRESERVED (Median 28, Ulnar 22, Radial 25) despite COMPLETE ANESTHESIA

Why? DRG sits OUTSIDE spinal canal. Avulsion = tear BETWEEN DRG and cord. Peripheral sensory axon still connected to DRG cell body = no Wallerian degeneration. Brain gets no signal, but the peripheral nerve still works.

- ALL motor responses ABSENT: motor neurons in anterior horn. Axons torn from cell bodies = Wallerian degeneration.

EMG -- WHY

- Every C5-T1 muscle: 3+ fibs, no MUAPs (completely denervated)

Cervical Paraspinals C5-T1: 3+ fibs. Posterior ramus branches at root level BEFORE plexus. Abnormal = root-level lesion.

The triad: preserved SNAPs + abnormal paraspinals + Horner's = root avulsion. Prognosis very poor.

CASE 4: Sudden Groin Pain and Buckling Knee

Femoral Neuropathy from Retroperitoneal Hematoma

68F on warfarin | INR 4.2 | Sudden onset after coughing | Hgb drop 12.5 to 9.8 | TIME-SENSITIVE

DIFFERENTIALS & RULE-OUTS

- L3-L4 Radiculopathy: adductors normal (obturator shares roots) + paraspinals normal
- Diabetic Amyotrophy: non-diabetic; amyotrophy = subacute bilateral
- Lumbar Plexopathy: would affect BOTH femoral + obturator. Adductors normal.
- Femoral at Inguinal Lig: iliopsoas WEAK = lesion is PROXIMAL to ligament

PHYSICAL EXAM -- WHY

- Hip held in flexion (psoas sign): hematoma in iliacus. Extension compresses it = pain.

Iliopsoas 2/5: femoral nerve branches to iliopsoas arise WITHIN iliac fossa, BEFORE inguinal ligament. Weak = lesion is in retroperitoneum.

- Quad 1/5: main femoral motor to quad is damaged

Adductors 5/5 (NORMAL): obturator nerve (L2-L4) is SEPARATE from femoral. Normal = not radic/plexopathy.

- Patellar reflex absent: femoral efferent limb destroyed

NCS -- WHY

- Sural/Superficial Fibular normal: sciatic territory separate from femoral

Saphenous ABSENT: sensory branch of femoral. Absent = post-ganglionic. In L3-L4 radic, saphenous = normal (DRG intact).

- Femoral motor ABSENT right, 6.5 left: confirms unilateral femoral nerve lesion

EMG -- WHY

Iliopsoas 2+ fibs: branches arise PROXIMAL to inguinal lig = lesion in iliac fossa/retroperitoneum

- All quad components 3+ fibs, no MUAPs: complete femoral denervation
- Adductor Longus NORMAL: obturator = separate. Rules out L2-L4 radic + plexopathy.
- Glut Med NORMAL: rules out plexopathy
- Paraspinals NORMAL: rules out radiculopathy

Treatment: reverse anticoagulation, CT scan, possible surgical decompression

CASE 5: Foot Drop + Hamstring Weakness After Surgery

Sciatic Nerve Injury at Hip (Post-THA)

67F | R total hip arthroplasty 6 weeks ago | Difficult anatomy | Awoke with foot drop + knee weakness

DIFFERENTIALS & RULE-OUTS

- Common Fibular at Fibular Head: tibial muscles (gastroc, tib post) + hamstrings would be NORMAL
- L5-S1 Radiculopathy: glutes + paraspinals would be abnormal
- Lumbosacral Plexopathy: glutes would be abnormal (arise from plexus PROXIMAL to sciatic)
- Cauda Equina: bilateral + saddle anesthesia + bowel/bladder

PHYSICAL EXAM -- WHY

Glut Med 5/5 + Glut Max 5/5 (BOTH NORMAL): sup/inf gluteal nerves arise from sacral plexus BEFORE sciatic forms. Normal = not plexus/root.

- Hamstrings 2/5: sciatic branches to hamstrings in thigh. Weak = lesion proximal to mid-thigh.
- TA 0/5, Gastroc 1/5: BOTH divisions affected. Peroneal worse than tibial (fewer fascicles, more vulnerable).
- Saphenous sensation preserved: femoral territory intact = isolated sciatic
- Achilles absent / Patellar normal: tibial arc destroyed, femoral arc intact

NCS -- WHY

- Sural + Superficial Fibular ABSENT: both sciatic divisions, post-ganglionic Wallerian degeneration
- Saphenous NORMAL: femoral territory separate
- Fibular motor ABSENT, Tibial motor 1.2 mV: peroneal division more severely affected
- H-reflex ABSENT right: S1 arc disrupted (both limbs through sciatic)

EMG -- WHY

- TA/EHL/Peroneus 3+ fibs, no MUAPs: peroneal division completely denervated
- Gastroc/Tib Post 2+ fibs, severely reduced: tibial division less severe

BFSH 2+ fibs: ONLY hamstring from peroneal division (branches in thigh). Abnormal = sciatic at hip. In fibular palsy at knee, BFSH = normal.

- BF Long Head + Semimembranosus 1+ fibs: tibial hamstrings less affected

Glut Max + Glut Med NORMAL: excludes plexopathy/root (branch before sciatic)

- Vastus Lat NORMAL: femoral territory intact
- Paraspinals NORMAL: not radiculopathy

Prognosis: 6-18 months partial recovery. Many have permanent foot drop.

QUICK COMPARISON -- ALL 5 CASES

	PTS	Klumpke	Root Avulsion	Retro Hematoma	Sciatic at Hip
Key SNAP	LABC abn	MAC abn	ALL preserved	Saphenous absent	Sural/SFib absent
Paraspinals	Normal	Normal	ABNORMAL	Normal	Normal
Key Muscle	Patchy multi-nerve	Thenar + hypothenar	All C5-T1	Iliopsoas	BFSH
Key Spared	Hand, pronator	Biceps, deltoid	None	Adductors	Glutes
Horner's	No	No	YES	No	No
Pre/Post Ganglionic	Post	Post	PRE	Post	Post

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